MSTAR PRACTICE WORKSHEET

Lesson 2-9: Describe Data by Mean Absolute Deviation

Grade 6 Mathematics · Standard 6.DS.6c

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

- **A** 4
- B −4
- C 11
- D 26

Deviation = 11 − 15 = −4. Absolute deviation = |−4| = 4.

Question 2 SELECTED RESPONSE — CORRECT: B

A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

- A 1
- **B** 3
- C 5
- D 15

Sum of absolute deviations = 3 + 1 + 5 + 2 + 4 = 15. MAD = 15 ÷ 5 = 3.

Question 3 SELECTED RESPONSE — CORRECT: A

The mean of a data set is 20. A value is 26. What is the absolute deviation?

- A 6
- B −6
- C 26
- D 46

> Deviation = 26 − 20 = 6. Absolute deviation = |6| = 6.

Question 4 SELECTED RESPONSE — CORRECT: D

Team A has MAD = 2.1 points. Team B has MAD = 6.8 points. Which team is more consistent?

- A Team B — higher MAD means better performance
- B Both are equally consistent
- C Cannot tell from MAD alone
- D Team A — lower MAD means scores are closer to the mean

> Lower MAD means less spread — Team A's scores stay closer to their average, making them more consistent.

Question 5 SELECTED RESPONSE — CORRECT: C

Two runners have the same average time of 60 seconds. Runner A's MAD is 1.2 seconds. Runner B's MAD is 5.8 seconds. Which runner is the coach more likely to pick for a relay race that needs a reliable time?

- A Runner B — higher MAD means faster potential
- B Either — they have the same average
- C Runner A — lower MAD means more consistent times
- D Neither — MAD doesn't matter for relay races

> Runner A's MAD of 1.2 means times are usually within 1.2 seconds of 60. Runner B's times vary more widely. For reliability, pick Runner A.

Question 6 SELECTED RESPONSE — CORRECT: B

A guard scored 2, 4, 6, 8 points in four scrimmages, so the mean is 5 points. The distances from the mean are 3, 1, 1, 3. What is the mean absolute deviation (MAD)?

A) 1

B) 2

C) 5

D) 8

> Add the distances: 3 + 1 + 1 + 3 = 8. Divide by the 4 games: 8 ÷ 4 = 2. The MAD is 2 points.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

A data set is 3, 5, 9, 11 with a mean of 7. What is the mean absolute deviation?

A) 2

B) 3

C) 7

D) 28

> Distances from the mean of 7 are 4, 2, 2, and 4. Their sum is 12, and 12 ÷ 4 = 3.

- **A:** Only the two inner distances were averaged. All four count: 4 + 2 + 2 + 4, divided by 4.
- **C:** That is the mean, which is where the distances are measured FROM. MAD averages those distances.
- **D:** That is the sum of the four data values. MAD averages the distances from the mean: 4, 2, 2, 4.

Part B — correct answer: C

Why are the DISTANCES from the mean used rather than the signed differences?

A) Distances are easier to add.

B) Negative numbers cannot be divided.

C) The signed differences always add to zero, so they would give a MAD of 0 for every data set.

D) The mean is always larger than every value.

> The mean is the balance point, so values above and below cancel exactly. Using absolute distances measures how far values sit from the center regardless of direction.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Deviations That Cancel

Sofia found the MAD of 3, 5, 9, 11 (mean 7). She used the signed differences −4, −2, 2, 4, added them to get 0, and concluded 'the MAD is 0, so there is no variability.'

Explain Sofia's misconception and give the correct MAD.

Model answer: MAD = (4 + 2 + 2 + 4) ÷ 4 = 3. The signed differences sum to zero for EVERY data set, because the mean balances the values above and below it. Taking absolute distances is what makes the measure meaningful.

2 points	Student explains that signed differences always sum to zero because the mean is the balance point, that absolute distances must be used, and computes (4 + 2 + 2 + 4) ÷ 4 = 3.
1 point	Student gives 3 without explaining why the signed differences cancel.
0 points	Student agrees with Sofia or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Two basketball teams both average 75 points per game. Team A's last 5 scores: 73, 76, 74, 77, 75. Team B's last 5 scores: 60, 90, 65, 85, 75. Calculate the MAD for each team and explain which team a coach would prefer if they need predictable scoring.

Model answer: Both teams average 75, so the mean cannot tell them apart — the MAD can. Team A's distances from 75 are 2, 1, 1, 2, 0, which total 6, so its MAD is 6 ÷ 5 = 1.2. Team B's distances are 15, 15, 10, 10, 0, which total 50, so its MAD is 50 ÷ 5 = 10. A coach who needs predictable scoring would choose Team A, because a MAD of 1.2 means its scores stay within about a point or two of 75 every game.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.